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4. Solving first order systems

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Numerically solving systems of first order ODEs

can obtain a solution to a
single differential equation

using the ODE45 function,
but how can we use ODE45
to solve a system of
differential equations,

like this one?

To do this, we'll first convert the system of ODE's into a new form, then we'll write

a function to compute derivative values, and lastly call ODE45, as we have done before.

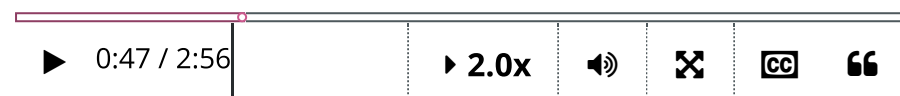
Let's get started.

The first step is to transform this system of differential equations into a single equation

for a single variable that we'll call 'x'.

The trick is to make 'x' a

$$\begin{aligned}\frac{dS}{dt} &= \frac{-2SI}{5000} \\ \frac{dI}{dt} &= \frac{2SI}{5000} - \frac{2I}{3} \\ \frac{dR}{dt} &= \frac{2I}{3}\end{aligned}$$



Video

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Solving higher order systems with ODE45

In general, to solve a n -dimensional system using ODE45 in MATLAB, the following syntax is used:

[t, x]=ode45(@ (t, x) odefun, tspan, x0)

outputs **inputs**

ODE45 requires that you provide the following three inputs:

1. **@(t,x) odefun:** **odefun** is a $n \times 1$ column vector $[f_1(t, \mathbf{x}); f_2(t, \mathbf{x}); \dots; f_n(t, \mathbf{x})]$ specifying the ODE to be solved. The notation **@(t,x)** is used to tell MATLAB that each f_i is a function of t and $\mathbf{x}$.
2. **tspan:** A 1×2 row vector $[t_0, t_{\text{end}}]$ specifying the start and end times between which to solve the system of DEs.
3. **x0:** A $n \times 1$ column vector specifying the initial conditions $\mathbf{x}(t_0)$.

In this course we will use the default settings of **ODE45** but additional inputs can be provided to control parameters such as the error tolerance and maximum step size. More details are can be found in the MATLAB documentation.

After it has finished numerically solving the DEs, **ODE45** will return two outputs:

1. $\mathbf{t}$: A column vector $[t_1, t_2, \dots, t_p]$ containing the times t_i at which the numerical solution has been computed.
2. $\mathbf{x}$: A matrix containing n columns $[\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n]$, where $\mathbf{x}_i$ is a column vector containing the values of the solution $\mathbf{x}$ at the times in the vector

containing the values of the solution $\mathbf{x}$, at the times in the vector $\mathbf{t}$.

Ungraded exploration

Use MATLAB Online to solve the system from the video for different (positive) initial conditions. Try plotting your solutions!

4. Solving first order systems

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